



NFISiS: New Perspectives on Fuzzy Inference Systems for Renewable Energy Forecasting

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Abstract

Deep learning (DL) models, despite their widespread adoption, present challenges related to extensive training times and a lack of interpretability. In contrast, fuzzy inference systems (FIS) offer a compelling balance between accuracy and transparency. This paper addresses the limitations of traditional Takagi–Sugeno–Kang (TSK) fuzzy models by proposing a novel Mamdani-based regressor derived from the recently introduced New Takagi–Sugeno–Kang (NTSK) framework. The resulting models are data-driven, enabling the number of fuzzy rules to be specified to manage the trade-off between accuracy and interpretability. To address the complexities of high-dimensional data, this work integrates both wrapper and ensemble techniques. A genetic algorithm (GA) is employed as a wrapper for feature selection, yielding genetically optimized model variants. Furthermore, ensemble models, namely the Random New Mamdani Regressor (R-NMR), Random New Takagi–Sugeno–Kang (R-NTSK), and Random Forest New Takagi–Sugeno–Kang (RF-NTSK), are introduced to improve robustness. The proposed methods are validated using photovoltaic (PV) energy forecasting datasets, a critical application area characterized by the intermittent nature of solar power. The fuzzy models, particularly those enhanced by genetic algorithms and ensembles, demonstrate highly competitive accuracy. For instance, the R-NTSK model achieved the lowest mean absolute percentage error (MAPE) among 29 competing methods on one of the evaluated datasets. Statistical analysis confirmed this robustness, revealing that some variants attained competitive performance with as few as a single fuzzy rule. These models frequently outperform both traditional machine learning and DL approaches while offering a simpler, more interpretable rule-based structure. To promote reproducibility, the models are provided in an open-source library named *nfisis* (<https://pypi.org/project/nfisis/>), and the datasets used in this study are publicly available at <https://doi.org/10.5281/zenodo.17179229>.

Keywords Mamdani · Feature selection · Regression · Time series · Renewable energy

1 Introduction

DL techniques have become increasingly popular due to technological advances and the growing complexity of databases. However, DL models are more intricate and susceptible to several issues, such as vanishing gradients, overfitting, long training times (Shrestha & Mahmood, 2019), and the inability to provide interpretable results due to their black-box nature (Wang et al., 2020; Rai, 2020). In contrast, fuzzy inference systems (FISs) represent a class of machine learning models

that combine accuracy with interpretability. These systems are primarily divided into two categories: the Mamdani-type (Mamdani, 1974) and the TSK-type (Takagi & Sugeno, 1985; Sugeno & Kang, 1988). These systems have been successfully applied in various fields, including finance, business, and management (Özkal et al., 2024; Nalbant, 2024; Nasehi, 2024), renewable energy (Sareen et al., 2023; Guerra et al., 2024), medicine (Abdalla et al., 2024), civil engineering (Emam et al., 2024), chemistry (Shaout & Brauchler, 2025), scheduling (Jailani et al., 2023), pattern recognition (Onoja et al., 2024), and fault detection (Yadav et al., 2024).

In this context, Alves et al. (2024a,b) proposed the NTSK to overcome several limitations identified in the literature of TSK models, including: (i) the absence of a standardized methodology for defining TSK rules; (ii) data-driven TSK models are often characterized by high complexity, numerous hyperparameters, and a reliance on hybrid approaches; (iii) a

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lack of direct control over the number of rules in data-driven methods; and (iv) the consequent component of TSK models exhibits poor explainability, as it is composed of polynomial functions. The NTSK is a data-driven fuzzy model designed for reduced complexity and enhanced interpretability. As highlighted by its authors, the model possesses several key advantages. First, the number of rules is a hyperparameter, granting the user explicit control over the trade-off between accuracy and interpretability. Second, the rules are defined based on the tangent of the target variable, a mechanism that improves the explainability of the model's consequent component. Finally, the approach is characterized by a minimal number of hyperparameters and low computational complexity, making it well-suited for various real-world applications. The simulations conducted in the original study demonstrated the model's superior performance, showing that it could outperform more complex models, including those from the DL paradigm.

On the other hand, Mamdani FIS share many of the same shortcomings as TSK-based models. These include a lack of direct control over the number of rules, a high number of hyperparameters, and no standard mechanism for rule creation. Consequently, researchers often resort to complex hybridization approaches to design the rule base. For instance, this has involved integrating Mamdani systems with neural networks (Anderson & Hall, 1999; Chai et al., 2009; Tung & Quek, 2009; Navarro-Almanza et al., 2022; Safari & Ghaemi, 2025; Akinsehinde et al., 2025), clustering algorithms like c-means (Gou et al., 2015; Uncu et al., 2004), or evolutionary computation, including GA and particle swarm optimization (PSO) (Gacto et al., 2014; Asadi, 2016; Kacimi et al., 2020; Ali et al., 2024).

Unlike the complex hybrid models, the NMR is an autonomous, data-driven model that constructs its rule-based structure directly from data. This approach provides several key advantages: i) it grants total control over the number of rules, which is specified as a single hyperparameter; ii) it automatically designs the fuzzy sets and rules without the need for hybridization; and iii) it significantly reduces the total number of hyperparameters. This process results in a simple yet accurate structure, making the model highly suitable for real-world implementation.

However, the increasing complexity and dimensionality of modern datasets present a critical challenge for this and other models: the trade-off between accuracy and interpretability. Datasets with hundreds of attributes may yield highly accurate predictions, but analyzing rules with hundreds of inputs is impractical for human understanding. The central question becomes which attributes to select to improve interpretability while maintaining accurate results. The solution lies in the application of feature selection techniques, which are broadly categorized into five paradigms: filter, wrapper, embedded, hybrid, and ensemble (Zebari et al., 2020).

These paradigms differ significantly in their approach. Filter methods, for instance, rely on statistical metrics to evaluate features independently of the learning algorithm; however, their main drawback is that they neglect the interaction between the selected feature subset and the model's predictive performance (Eesa et al., 2015). In contrast, wrapper methods evaluate feature subsets by using the performance of a specific learning algorithm as the primary selection criterion. As an exhaustive search of all feature combinations is computationally infeasible, these methods employ heuristic or metaheuristic algorithms, such as randomized search (Mao & Yang, 2019), GA (Sohail, 2023), and ant colony optimization (Forsati et al., 2014). In embedded methods, the feature selection process is integrated directly into the training of the learning algorithm itself, typically by optimizing internal model parameters. Common examples of models with embedded feature selection capabilities include decision trees, random forests (RF), and gradient boosting (Guyon & Elisseeff, 2003). Hybrid methods combine techniques from the aforementioned categories in a multi-step process (Hsu et al., 2011). Finally, ensemble learning leverages multiple weak learning algorithms, combining their outputs to achieve greater predictive performance than any single constituent model (Bolón-Canedo & Alonso-Betanzos, 2019; Dong et al., 2020).

To address this challenge, this paper integrates wrapper and ensemble feature selection concepts with the NMR and NTSK frameworks. This integration yields a collection of new fuzzy inference system (NFISiS) designed to handle high-dimensional datasets, optimize predictive performance, and enhance interpretability. First, a wrapper technique that employs a GA to search for the optimal attribute combination is introduced, resulting in the Genetic NMR (GEN-NMR) and Genetic NTSK (GEN-NTSK) models. Second, this work introduces three novel ensemble models: the Random NMR (R-NMR), the Random NTSK (R-NTSK), and the Random Forest NTSK (RF-NTSK). The R-NMR and R-NTSK are homogeneous ensembles, while the RF-NTSK is a heterogeneous ensemble combining the R-NTSK and a conventional RF.

It is crucial to distinguish this work from the hybrid models cited previously. While those approaches use evolutionary algorithms or ensembles to design the fuzzy rules themselves, our proposed models apply these techniques exclusively for feature selection. This preserves the autonomous, data-driven rule and fuzzy set generation of the base NMR and NTSK models, thereby directly targeting the challenge of high dimensionality without sacrificing the models' core simplicity.

The proposed models are validated on renewable energy time series datasets, specifically focusing on photovoltaic (PV) energy forecasting. This application was selected due to its inherent complexity and significant real-world relevance.

PV energy is poised to become a primary global energy source, owing to its abundance, affordability, and scalability (Alcañiz et al., 2023). A key challenge, however, is the intermittent nature of PV generation, which is highly dependent on fluctuating weather conditions (Notton et al., 2018). Consequently, accurately forecasting the power output of PV plants is a critical task. Such forecasts enable grid operators, plant managers, and energy markets to anticipate and mitigate the variability inherent in solar power. Accurate predictions are essential for optimizing energy production, ensuring power grid stability, reducing operational costs, and facilitating efficient resource planning, thereby contributing to a more sustainable energy landscape. Moreover, the chaotic dynamics and uncertainties of PV systems often render traditional physical models inadequate for precise forecasting, motivating the widespread adoption of machine learning approaches for this task (Mayer & Gróf, 2021).

The critical importance of PV energy control motivated extensive research into developing accurate predictive models. Wan et al. (2015); Barbieri et al. (2017); Das et al. (2018); Alcañiz et al. (2023) presented an extensive literature review on this topic. Munsif et al. (2023) proposed the convolutional transformer-based network (CT-NET) for power forecasting. Jailani et al. (2023) implements LSTM for solar energy. Artificial neural networks (ANN) (El Haji et al., 2025) and ANN with feature selection (Du et al., 2025) were also discussed. Sharma et al. (2023) utilized a Levenberg–Marquardt artificial neural network (LM-ANN) that uses the gradient descent (GD) optimization technique for solar energy forecasting. Among the advantages of using ANNs, the following can be highlighted: (i) self-adaptive ability; (ii) fault tolerance; (iii) robustness; and (iv) strong inference capabilities (Yang et al., 2014). However, due to the architecture of ANNs, they can present increased complexity. Support vector machines (SVM) have also been explored for PV energy forecasting (Preda et al., 2018), praised for their non-linear modeling capacity and independence from prior knowledge (Li et al., 2016). Numerous hybrid approaches using SVM have emerged, such as PSO-SVM (Eseye et al., 2018), GASVM (VanDeventer et al., 2019), and SVM with an improved ant colony optimization (ACO-SVM). Fuzzy models were largely applied to the power predictions of PV systems (Liu et al., 2017, 2021; Vijayalakshmi et al., 2024; Salameh et al., 2025; Jalalifar et al., 2025). A remarkable advantage of fuzzy models is their ability to handle uncertainties in real-world data (Ghofrani et al., 2016). Still discussing fuzzy models, ANFIS usually presents errors substantially lower than those of fuzzy models. However, the computational complexity also increases, especially compared to SVM (Das et al., 2018; Law et al., 2014).

The remainder of this paper is organized as follows. Section 2 details the theoretical background of the NTSK model. Section 3 introduces the proposed models and describes their

mathematical formalism. Section 4 presents the experimental setup, simulation results, and a comprehensive discussion. Finally, Sect. 5 summarizes the key findings of this work and outlines directions for future research.

2 Background Knowledge on NTSK

NTSK is a data-driven FIS that automatically forms TSK rules based on the desired number of fuzzy rules informed by the user. The model adheres to the canonical structure of a TSK system, where each rule $\mathcal{R}_i$ is expressed as:

$$\mathcal{R}_i : \quad \text{IF } \underbrace{x \text{ is } \mathcal{A}_i}_{\text{Antecedent}} \quad \text{THEN } \underbrace{y_i = f_i(x, \theta_i)}_{\text{Consequent}} \quad (1)$$

where $\mathcal{R}_i$ is the i th fuzzy rule, $i = 1, 2, \dots, R_{\max}$, $R_{\max}$ is the total number of fuzzy rules, $x = [x_1, \dots, x_p]^T \in \mathbb{R}^p$ is the input vector, p is the number of attributes in the input vector, y_i is the target value, θ_i is the parameters of the consequent (polynomial functions), and $y_i = f_i(x, \theta_i)$ express that the target value is a function of the input vector and the parameters of the consequent. In this structure, the antecedent consists of the multivariate fuzzy set $\mathcal{A}_i$, while the consequent y_i is a crisp function of the input vector x and a set of parameters θ_i . The rule implies that if an input vector x has a high degree of membership in the fuzzy set $\mathcal{A}_i$, its corresponding output is determined by the function $f_i(x, \theta_i)$.

The primary novelty of the NTSK model lies in its rule induction mechanism, which constitutes the core of its training process. In contrast to conventional TSK methods that typically first define antecedent fuzzy sets and subsequently establish the rule base, the NTSK algorithm reverses this procedure. The training process commences by partitioning the output space of the target variable into a set of intervals. Each resulting interval is then used to establish a distinct fuzzy rule. Only after establishing this rule base does the algorithm proceed to define the antecedent fuzzy sets ($\mathcal{A}_i$) that map the input space to these consequent rules. Upon completion of training, the model is ready for the inference phase. Accordingly, the remainder of this section provides a detailed overview of both the training and inference procedures for the NTSK model.

2.1 Defining NTSK Rules: Training Phase

The training process for the NTSK model consists of six primary steps, which are summarized below and detailed throughout this subsection.

- **Step 1—Specify the Number of Rules:** The user specifies the desired number of rules ($R_{\max}$).

- **Step 2—Compute Target Value Variations:** The first-order difference is calculated for consecutive target values.
- **Step 3—Define Variation Intervals:** Equally spaced intervals are created based on the range of target value variations and $R_{\max}$.
- **Step 4—Assign Samples to Rules:** Each data sample is assigned to a rule based on the value or the variation of the sample and the variation interval of each rule.
- **Step 5—Compute Antecedent Parameters:** The parameters for the antecedent fuzzy sets are determined (e.g., mean and standard deviation for Gaussian membership functions).
- **Step 6—Compute Consequent Parameters:** The consequent parameters (θ) are estimated using an adaptive filtering method.

The training begins with the user specifying the maximum number of rules, $R_{\max}$. This hyperparameter provides direct control over the number of rules a notable advantage compared to many data-driven FIS approaches. Next, the model analyzes the dynamics of the target variable by computing its first-order difference, which captures the local rate of change between consecutive samples. This is calculated as:

$$\tan(y^k) = \frac{y^{k+1} - y^k}{(k+1) - k} = y^{k+1} - y^k = \Delta y^k \quad (2)$$

where $\tan(y^k)$ is the tangent of the k th sample, with $k = [1, \dots, N-1]$, where N is the total number of samples. It is intuitive to show that the tangent is equivalent to compute the first-order difference in this context of time series since $\tan(y^k) = y^{k+1} - y^k = \Delta y^k$. Consequently, tangent and first-order difference are used interchangeably and designated by Δy^k for convenience.

The primary motivation for computing the tangent is to capture the local trends and rates of change in the target variable, as illustrated for a sample point in a time series in Fig. 1. After calculating this variation (Δy^k) for all applicable samples, the model partitions the range of these values into distinct intervals. This partitioning serves to group samples that exhibit similar magnitudes of variation. Each interval is defined as:

$$\text{Range}_i = [\underline{m}_\Delta + (i-1)IS, \underline{m}_\Delta + (i)IS] \quad (3)$$

where Range_i represents the intervals' limits for the tangent of the i th rule, for $i = [1, 2, \dots, R_{\max}]$, $R_{\max}$ is the appointed number of rules, $\bar{m}_\Delta = \max(\Delta y^k)$ is the maximum tangent value, $\underline{m}_\Delta = \min(\Delta y^k)$ is the minimum tangent value, and

the interval size (IS) is computed as:

$$IS = \frac{\bar{m}_\Delta - \underline{m}_\Delta}{R_{\max}} \quad (4)$$

Subsequently, the model assigns each sample to its corresponding rule according to the criterion:

$$i_{sk} = \begin{cases} \lfloor \frac{\Delta y^k - \underline{m}_\Delta}{IS} \rfloor + 1, & \text{if } \Delta y^k < \bar{m}_\Delta \\ R_{\max}, & \text{otherwise} \end{cases} \quad (5)$$

where i_{sk} is the rule that the k th sample will be included and $\lfloor \cdot \rfloor$ represents the floor function.

The final training step involves computing the consequent parameters θ . As demonstrated by Alves et al. (2024b), this can be accomplished using one of two adaptive filters: recursive least squares (RLS) or weighted recursive least squares (wRLS). The choice of filter has significant implications for model behavior. The RLS approach estimates a single, global set of parameters (θ) for all rules, meaning model performance is decoupled from the number of rules, which primarily affects interpretability. In contrast, the wRLS approach estimates a unique set of parameters for each rule, making model performance directly dependent on the chosen value of $R_{\max}$.

A flowchart summarizing the complete training phase of the NTSK model is presented in Fig. 2.

2.2 The Inference Process of NTSK: Test Phase

During the inference or test phase, the trained NTSK model predicts the output for unseen input vector (x^k). The estimated output ($\hat{y}^k$) is calculated as the weighted sum of the consequent for all fuzzy rules:

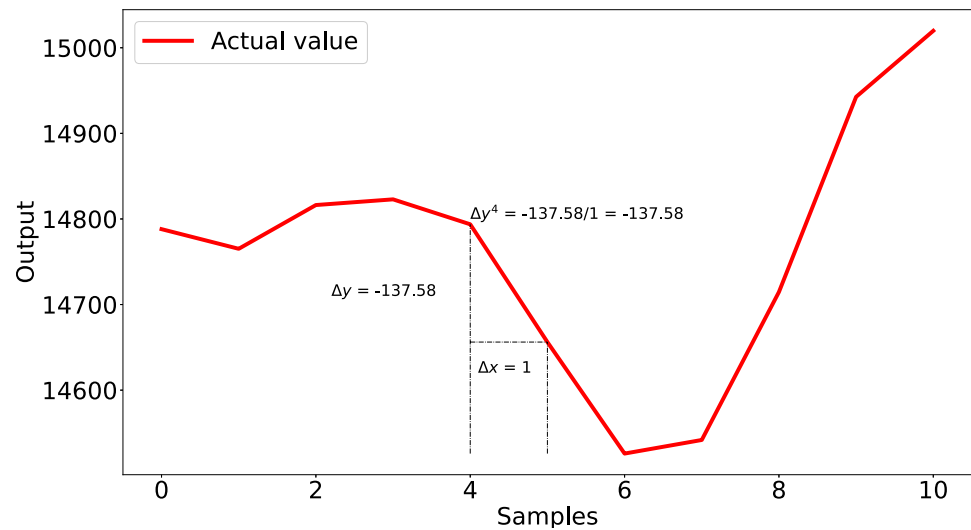
$$\hat{y}^k = \sum_{i=1}^{R_{\max}} w_i (x^k)^T \theta_i \quad (6)$$

where $x^k = [1, x^T]^T$ is the input vector with the bias and w_i is the normalized firing degree of the i th rule as:

$$w_i = \frac{\prod_{j=1}^p \exp \left[-\frac{1}{2} \frac{(x_j^k - v_{i,j})^2}{\sigma_{i,j}^2} \right]}{\sum_{i=1}^{R_{\max}} \left(\prod_{j=1}^p \exp \left[-\frac{1}{2} \frac{(x_j^k - v_{i,j})^2}{\sigma_{i,j}^2} \right] \right)} \quad (7)$$

where $v_i = [v_{i,1}, \dots, v_{i,p}]^T$ with $v_{i,j}$ consisting of the mean of the j th attribute for the i th rule, p is the dimension of the inputs (number of attributes for each sample), x_j^k is the k th input vector for the j th attribute, and $\sigma_i = [\sigma_{i,1}, \dots, \sigma_{i,p}]^T$

Fig. 1 Illustration of Δy^k calculation



with $\sigma_{i,j}$ consisting of the standard deviation of the j th attribute for the i th rule.

3 The Proposed Models

This paper presents a significant extension of the NTSK framework through three primary contributions. The first is the introduction of the NMR, a novel model that adapts the core mechanisms of NTSK to a Mamdani fuzzy architecture. The second contribution details the integration of a GA with both the NTSK and NMR models for the purpose of automated feature selection. The third and final contribution is the development of several ensemble fuzzy models, creating a robust framework designed for time series forecasting. To enhance reproducibility and facilitate further research, all proposed models are implemented in the open-source Python package, NFISiS (Alves & de Aguiar, 2025). The following subsections describe each of these contributions in detail.

3.1 NMR Model

The NMR model, like its NTSK counterpart, is a data-driven FIS that constructs its rule base by partitioning the output space. The primary distinction between the two models lies in the variable used for this partitioning. The NTSK model utilizes the first-order difference of the target values to define its intervals, whereas the NMR model defines them based directly on the target values themselves.

3.1.1 Defining NMR Rules: Training Phase

The training process that consists of the creation of NMR rules summarized the five steps presented below:

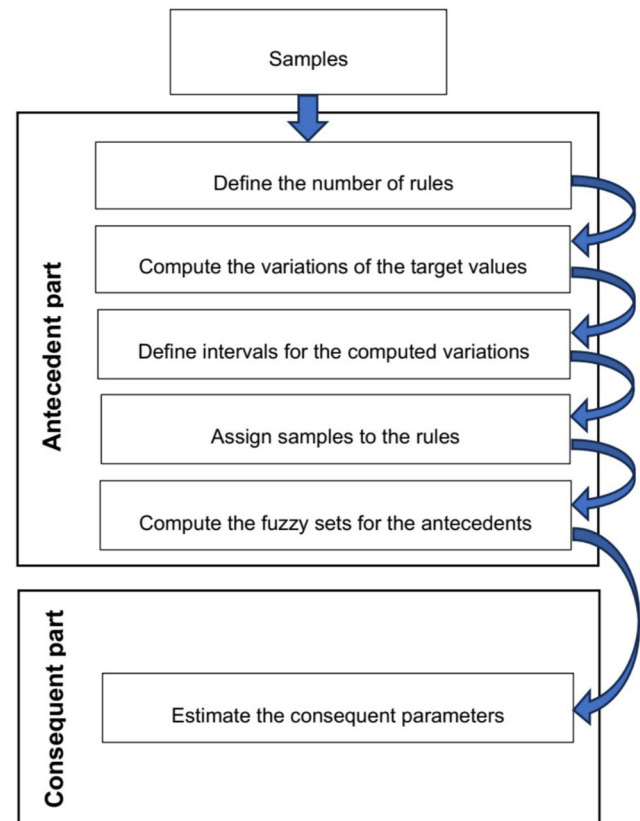


Fig. 2 Flowchart to illustrate the learning phase for NTSK

Step 1—Define the Number of Rules: The user specifies the desired number of rules ($R_{\max}$).

Step 2—Partition the Output Space: The model creates equally spaced intervals across the range of the target variable.

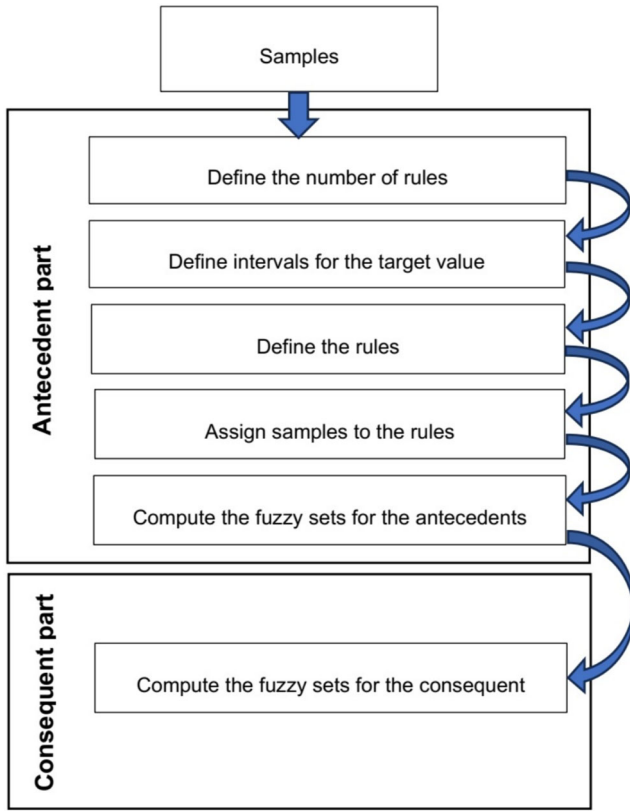


Fig. 3 Flowchart of the learning phase for NMR

Step 3—Establish Consequent Rule Definitions: The intervals informs to which rule each sample will be included by comparing the target value with each interval region.

Step 4—Assign Samples to Rules: Each data sample is assigned to the rule corresponding to its target value's interval.

Step 5—Compute Fuzzy Set Parameters: The parameters for the antecedent and consequent membership functions are calculated from the partitioned data.

Figure 3 illustrates the steps in the training phase of NMR. Once trained, the model is ready for inference.

The procedure begins with the user specifying the number of rules as $R_{\max}$, which is a key hyperparameter controlling model granularity. The algorithm then computes the size of each output interval (IS) based on the amplitude of the target variable's range as:

$$IS = \frac{\bar{y} - \underline{y}}{R_{\max}} \quad (8)$$

where $\bar{y}$ and $\underline{y}$ are the maximum and minimum values of the target variable in the training data, respectively. Then, IS is used to compute each interval for the consequent part of all rules, as:

$$Range_i = [\underline{y} + (i - 1)IS, \underline{y} + (i)IS] \quad (9)$$

where $Range_i$ represents the interval of the consequent part of the i th rule and $i = 1, 2, \dots, R_{\max}$.

Once the consequent intervals are defined, each training sample is assigned to its corresponding rule by comparing its target value to these intervals. The index of the rule assigned to the k th sample is determined by:

$$i_{Sk} = \begin{cases} \lfloor \frac{y^k - \underline{y}}{IS} \rfloor + 1, & \text{if } y^k < \bar{y} \\ R_{\max}, & \text{otherwise} \end{cases} \quad (10)$$

where i_{Sk} is the rule in which the k th sample will be included and $\lfloor \cdot \rfloor$ represents the floor function.

The final step in the rule definition process is to compute the parameters for the antecedent and consequent fuzzy sets. This work employs Gaussian membership functions, defined by the equation:

$$\mu_A(x) = \exp \left[-\frac{1}{2} \left(\frac{x - v}{\sigma} \right)^2 \right] \quad (11)$$

where v represents the mean of the fuzzy set and σ is the standard deviation.

For each rule i , the parameters of these functions are derived from the subset of data samples assigned to it:

- **Antecedent Fuzzy Sets:** For each input attribute j of rule i , the mean ($v_{i,j}$) and standard deviation ($\sigma_{i,j}$) are calculated from the subset of input data whose samples were assigned to rule i .
- **Consequent Fuzzy Sets:** For the consequent of rule i , the mean ($v_{i,y}$) and standard deviation ($\sigma_{i,y}$) are calculated from the subset of input data whose samples were assigned to rule i .

Gaussian membership functions are employed due to their advantageous properties: they are smooth, universally nonzero, and effective for representing linguistic variables (Azimi & Miar-Naimi, 2020). The complete training procedure is summarized in the pseudo-code presented in Algorithm 1.

3.1.2 The Inference Process for NMR: Test Phase

During the inference phase, the NMR model first computes the normalized firing degree (w_i) for each rule according to (7). The final output ($\hat{y}^k$) is then estimated as:

$$\hat{y}^k = \sum_{i=1}^{R_{\max}} v_{i,\text{output}} \times w_i \quad (12)$$

where $v_{i,\text{output}}$ is mean of the consequent fuzzy set for the i th rule.

Algorithm 1: NMR for the training phase

Input: x, y
1 Initialization: $R_{\max}$
 Compute the interval size:
 Equation
2 (8)
3 for $i = 1, 2, \dots, R_{\max}$ **do**
 | Compute the interval:
 | Equation
4 | (9)
5 for $k = 1, 2, \dots, N - 1$ **do**
 | Compute the rule which the sample will be assigned
6 | i_{sk} : Equation (10)
 | Assign the samples using
7 i_{sk}
8 for $i = 1, 2, \dots, R_{\max}$ **do**
 | Compute the parameters of the fuzzy sets

The complete inference procedure for the NMR model is summarized in the pseudo-code presented in Algorithm 2.

Algorithm 2: NMR for the test phase

Input: x
Output: $\hat{y}$
1 for $k = 1, 2, \dots$ **do**
 | Compute the normalized firing degree:
 | Equation
2 | (7)
 | Compute the output:
 | Equations
3 | (12)

To clearly delineate the main differences between the NMR and NTSK models, a detailed side-by-side comparison is provided in Table 1.

3.2 GA for Feature Selection

GA is employed as a wrapper method for feature selection, leading to the development of two new models: GEN-NMR and GEN-NTSK. This approach aims to identify the most informative subset of input features. In this framework, each candidate feature subset is encoded as a chromosome, a binary vector whose length is equal to the total number of input attributes. A value of '1' in the vector signifies that the corresponding feature is selected, while a '0' signifies its exclusion. A constraint is enforced to ensure that each chromosome contains at least one selected feature, thereby preventing null feature sets.

The fitness of each chromosome is evaluated based on the performance of the corresponding fuzzy model (NTSK or NMR), which is trained and validated using only the feature subset encoded by that chromosome. Through the iterative

Table 1 Main differences between NMR and NTSK

Aspect	NMR	NTSK
Rule induction	Partitions the space of the target variable (y)	Partitions the space of the target variable's first-order difference (Δy)
Rule consequent	Fuzzy sets (linguistic terms)	Polynomial functions (mathematical expressions)
Output computation	Requires defuzzification to yield a crisp output	Uses a weighted average of the crisp outputs from each rule
Consequent parameters	Fuzzy set parameters (e.g., mean, std. dev.) derived from data partitions	Function parameters (θ_i) estimated via adaptive algorithms (e.g., RLS)
Computational load	Generally lower due to simpler consequent structure	Typically higher due to parameter estimation for consequent functions
Key strength	More interpretable ("transparent") due to linguistic rule consequents	Often yields higher predictive accuracy due to the power of functional consequents

application of standard evolutionary operators—selection, crossover, and mutation—the GA refines a population of chromosomes to find the feature combination that maximizes fitness (i.e., minimizes prediction error). The final output of this process is the optimal feature subset identified. The key hyperparameters governing the GA's operation, along with their default values, are summarized in Table 2.

The procedural implementation of the GA for feature selection is outlined in the pseudo-code in Algorithm 3.

Algorithm 3: GA for feature selection

Input: x, y
Output: attributes_vector
 Internally separate the samples into train and test.
 Initialize the vector of binary elements.
 Generate a subset for each vector.
 Call the fuzzy model and evaluate the fitness of the initial population.
 1 **for** $generation = 1, 2, \dots, n_generations$ **do**
 Select the best individuals to constitute the parents.
 Apply reproduction to generate a new population.
 Call the fuzzy model and evaluate the fitness of the generated population.
 Select the individual with the best fitness value as the winner.

3.3 Ensemble Fuzzy

As the final contribution, this work develops three distinct fuzzy ensemble models to enhance predictive accuracy and robustness: R-NTSK, R-NMR, and RF-NTSK. The efficacy of ensemble methods is found on two key principles: the diversity of the base learners and their individual predictive performance.

The R-NTSK and R-NMR models are homogeneous ensembles based on the Random Subspace method. For each ensemble, a total of Z base models (NTSK or NMR) are trained. Each base model is trained on a different, randomly selected subset of the available input features. The final prediction of the ensemble is the average of the outputs from all Z individual models.

The RF-NTSK model is a heterogeneous meta-ensemble that combines the predictions of two distinct models: a RF and the previously described R-NTSK ensemble. The final output is a weighted average of the predictions from these two models, where the weights are inversely proportional to each model's error on the training data. This allows the meta-ensemble to favor the prediction of the more accurate predictor. This aggregation is formulated as:

$$\hat{y}^k = \hat{y}_{RF}^k \frac{\epsilon_{R-NTSK}}{\epsilon_{RF} + \epsilon_{R-NTSK}} + \hat{y}_{R-NTSK}^k \frac{\epsilon_{RF}}{\epsilon_{RF} + \epsilon_{R-NTSK}} \quad (13)$$

where ϵ_{RF} and ϵ_{R-NTSK} are the estimated error of RF and R-NTSK during the training phase, respectively, and $\hat{y}_{RF}^k$

and $\hat{y}_{R-NTSK}^k$ the output of RF and R-NTSK, respectively. It can be seen that the output is a weighted average of the output of RF and R-NTSK where the weights are inversely proportional to the errors.

The general procedure for constructing the random subspace ensembles is outlined in the pseudo-code in Algorithm 4.

Algorithm 4: The ensemble fuzzy algorithm

Input: x, y
Output: $\hat{y}$
 1 **for** $i = 1, 2, \dots, n_estimators$ **do**
 2 **for** $j = 1, 2, \dots, n_trials$ **do**
 Generate a subset of the original dataset.
 Separate 75% of the subset for train and the rest for test.
 Evaluate the fitness for each subset.
 Select the best model to compose the list of estimators.

4 Experimental Results and Discussion

This section first details the performance metrics employed for model evaluation. Subsequently, it presents the datasets used for the simulations. Finally, it reports the results obtained from the experimental simulations and provides a comprehensive discussion of the findings and their implications.

4.1 Evaluation Indicators

The model's performance is assessed using the normalized root mean square error (NRMSE), the non-dimensional index error (NDEI), and the mean absolute percentage error (MAPE), defined, respectively, by:

$$NRMSE = \frac{RMSE}{\bar{y} - \underline{y}} \quad (14)$$

$$NDEI = \frac{RMSE}{std([y^1, \dots, y^T])} \quad (15)$$

$$MAPE = \frac{1}{T} \sum_{k=1}^T \frac{\|y^k - \hat{y}^k\|}{y^k} \quad (16)$$

where y^k is the k th actual value, $\hat{y}^k$ is the k th predicted value, $\bar{y}$ is the maximum value for y , $\underline{y}$ is the minimum value for y , T is the sample size, $std()$ is the standard deviation function,

Table 2 GA hyperparameters and default values

Hyperparameter	Description	Default value
num_generations	The total number of evolutionary generations the algorithm will run	10
num_parents_mating	The number of elite individuals selected to produce offspring for the next generation	5
sol_per_pop	The total number of solutions (chromosomes) maintained in the population for each generation	10
error_metric	The fitness function used to evaluate solutions. The GA's objective is to find a feature set that minimizes this error. Options include RMSE, NRMSE, NDEI, MAE, and MAPE	'RMSE'

Table 3 Characteristics of the PV panels implemented in the simulations

Characteristics	Alice 1A	Alice 38	Yulara 1	Yulara 5
Manufacturer	Trina	Q CELLS	–	–
Array rating	10.5 kW	5.9 kW	1058.4 kW	105.9 kW
PV technology	mono-Si	mono-Si	poly-Si	mono-Si
Array structure tracker	Dual Axis	Ground Mount	Ground Mount	Roof Mount
Installed	2009	2017	2016	2016

Table 4 Statistical tests of the PV energy datasets

Test	Alice 1A	Alice 38	Yulara 1	Yulara 5
Shapiro–Wilk	5.0×10^{-17}	1.4×10^{-27}	5.4×10^{-11}	0.37
ADF	8.5×10^{-8}	1.0×10^{-20}	3.7×10^{-20}	8.3×10^{-4}

and RMSE is the root mean square error given by:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{k=1}^T (y^k - \hat{y}^k)^2} \quad (17)$$

Each metric provides a different perspective on the model's accuracy. RMSE is widely used because its quadratic scoring rule heavily penalizes large errors and is suitable for gradient-based optimization. To provide a scale-independent evaluation, two normalized versions of RMSE are reported. The NRMSE normalizes the error by the range of the target data, which is useful for comparing performance across datasets with different scales. The NDEI normalizes the RMSE by the standard deviation of the data, providing insight into whether the model's error is smaller than the data's natural variability. Finally, while the previous metrics focus on absolute error, MAPE is included to evaluate performance based on relative percentage error, which can be more intu-

itive. However, it is important to note that MAPE is not suitable for time series containing zero values due to the division-by-zero issue. The error results are reported using tables, with the results grouped into four groups. The first group comprises the classical models, which are the majority of the scikit-learn library. The second group represents the DL models. The third group consists of the evolving fuzzy systems (eFS). Finally, the last group contains the models discussed in this paper. For rule-based models, the total number of final rules is also reported.

To formally assess the statistical significance of performance differences, the modified Diebold–Mariano (MDM) test, as proposed by Harvey et al. (1997), is employed. The null hypothesis of this test posits that the two models under comparison exhibit equal predictive accuracy. The hypothesis is tested at a significance level of $\alpha = 0.05$. In the results tables, a p-value below this threshold leads to the rejection of. Such a rejection indicates that the observed difference in

forecast accuracy is statistically significant, supporting that the model in the column has a lower prediction error than the model in the row. The analysis was conducted using the dieboldmariano Python library.

4.2 Datasets

The experiments in this study are conducted using PV energy datasets from two large-scale power plants located in Australia: the Desert Knowledge Australia Solar Centre (DKASC) in Alice Springs and the Yulara Solar System

The DKASC is a comprehensive solar energy facility that has been in operation since 2008. It features a wide variety of PV technologies with different ages, manufacturers, and configurations, providing a rich source of data for research. This work utilizes data from this facility to perform one-step-ahead daily power forecasting. The set of predictors for the Alice Springs models includes meteorological and electrical variables such as: humidity [%], diffuse radiation [W/m^2], radiation [W/m^2], diffuse tilted [W/m^2], accumulated energy [kWh], rainfall [mm], wind direction [$^\circ$], temperature [$^\circ\text{C}$], radiation global [W/m^2], tilted [W/m^2], energy [kWh], global radiation [W/m^2], power [kW], and the current [A].

The Yulara Solar System, installed in 2014, operates a 1.8 MW solar PV plant. Two specific PV arrays from this site, Yulara 1 and Yulara 5, were selected for this study. The prediction task is again one-step-ahead daily power forecasting, using a set of predictors including: air pressure [not specified], wind direction [$^\circ$], rainfall [mm], hail [not specified], wind speed [m/s], accumulated energy [kWh], max wind speed [m/s], pyranometer [not specified], global radiation [W/m^2], temperature [$^\circ\text{C}$], temperature probe 1 [$^\circ\text{C}$], temperature probe 2 [$^\circ\text{C}$], energy [kWh], power [kW], and the current [A].

The raw data for all datasets were recorded at a 5-minute interval. A preprocessing step was performed to resample the time series to a daily frequency and handle any null values. The analysis period for this study covers data from January 2021 to December 2022. For each dataset, the data were chronologically partitioned into three non-overlapping sets: 60% for training, 20% for validation, and 20% for testing. The training set is used to fit the model parameters, the validation set is used for hyperparameter tuning, and all final evaluation metrics are computed on the unseen test set to ensure an unbiased assessment of performance.

Detailed specifications for the selected PV panels are presented in Table 3. Further information is available on the DKASC website: <https://dkasolarcentre.com.au/>. To promote reproducibility, the final, preprocessed datasets used in this study are publicly available at <https://doi.org/10.5281/zenodo.17179229> (Alves, 2025).

To analyze the underlying properties of the time series, two statistical tests were conducted: the Shapiro–Wilk test

Table 5 Simulations' results for the Alice 1A dataset

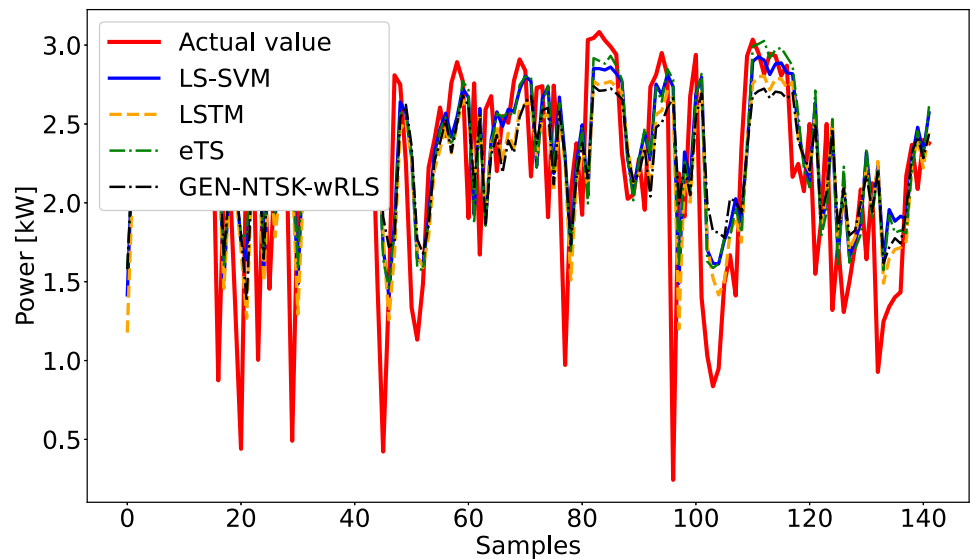
Model	NRMSE	NDEI	MAPE	Rules
<i>Classical Models</i>				
KNN	0.22656	1.00709	0.40166	–
Regression Tree	0.21325	0.94790	0.37171	–
Random Forest	0.21480	0.95481	0.36574	–
SVM	0.20714	0.92075	0.35383	–
LS-SVM	0.20669	0.91876	0.35509	–
GBM	0.21849	0.97119	0.36887	–
XGBoost	0.21538	0.95736	0.37573	–
LGBM	0.21320	0.94771	0.37708	–
<i>DL</i>				
MLP	0.26037	1.15734	0.46606	–
CNN	0.22639	1.00634	0.40184	–
RNN	0.20881	0.92818	0.36124	–
LSTM	0.20776	0.92350	0.35657	–
GRU	0.26014	1.15636	0.39765	–
WaveNet	0.25401	1.12910	0.45572	–
<i>eFS</i>				
eTS	0.20639	0.91742	0.35508	4
Simpl_eTS	0.33692	1.49764	0.43992	56
exTS	0.20861	0.92729	0.36044	3
ePL	0.20953	0.93136	0.35760	1
eMG	0.40435	1.79734	0.56272	140
ePL+	0.20757	0.92266	0.34947	3
ePL-KRLS-DISCO	0.29263	1.30077	0.46094	33
<i>Discussed Models</i>				
NMR	0.24337	1.08178	0.38902	16
NTSK (RLS)	0.24953	1.10918	0.43855	1
NTSK (wRLS)	0.20957	0.93154	0.35129	4
GEN-NMR	0.23421	1.04108	0.38867	17
GEN-NTSK (RLS)	0.22072	0.98112	0.36194	1
GEN-NTSK (wRLS)	0.20711	0.92061	0.36308	19
R-NMR	0.23220	1.03216	0.36129	–
R-NTSK	0.21431	0.95261	0.35784	–
RF-NTSK	0.21177	0.94132	0.35908	–

Bold values denote the lowest error values observed for each class of model

for normality (Shapiro & Wilk, 1965) and the augmented Dickey–Fuller (ADF) test for stationarity (Dickey & Fuller, 1981). For the Shapiro–Wilk test, a p-value below 0.05 indicates that the data are not normally distributed. For the ADF test, a p-value below 0.05 suggests that the series is stationary. The results of these tests are shown in Table 4. The p-values suggest that only the Yulara 5 dataset originates from a normal distribution and that all four datasets are stationary.

Table 6 Statistical test for Alice 1A

Model	NMR	NTSK (RLS)	NTSK (wRLS)	GEN-NMR	GEN-NTSK (RLS)	GEN-NTSK (wRLS)	R-NMR	R-NTSK	RF-NTSK
KNN	0.85	1.00	0.07	0.70	0.27	0.00	0.65	0.10	0.05
Regression Tree	0.99	1.00	0.29	0.95	0.82	0.14	0.97	0.56	0.37
Random Forest	0.98	1.00	0.18	0.93	0.80	0.07	0.96	0.47	0.14
SVM	0.99	1.00	0.74	0.98	0.99	0.50	1.00	0.99	0.94
LS-SVM	0.99	1.00	0.74	0.98	1.00	0.55	1.00	0.99	0.97
GBM	0.96	1.00	0.14	0.88	0.60	0.06	0.88	0.31	0.13
XGBoost	0.97	1.00	0.19	0.92	0.77	0.04	0.93	0.43	0.18
LGBM	0.99	1.00	0.30	0.96	0.83	0.05	0.95	0.57	0.37
MLP	0.15	0.18	0.00	0.04	0.00	0.00	0.05	0.00	0.00
CNN	0.97	0.95	0.06	0.86	0.33	0.01	0.67	0.14	0.09
RNN	1.00	1.00	0.55	0.99	0.96	0.37	1.00	0.87	0.69
LSTM	1.00	1.00	0.67	0.98	0.99	0.44	1.00	0.98	0.82
GRU	0.00	0.30	0.00	0.00	0.01	0.00	0.04	0.00	0.00
WaveNet	0.24	0.37	0.00	0.07	0.02	0.00	0.12	0.00	0.00
eTS	0.99	1.00	0.75	0.98	0.99	0.57	1.00	0.97	0.96
Simpl_eTS	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
exTS	0.99	1.00	0.57	0.97	0.97	0.37	0.98	0.88	0.81
ePL	0.99	1.00	0.50	0.96	0.99	0.28	0.99	0.94	0.78
eMG	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ePL+	0.99	1.00	0.74	0.98	0.99	0.46	1.00	0.98	0.91
ePL-KRLS-DISCO	0.03	0.03	0.00	0.01	0.00	0.00	0.00	0.00	0.00

Fig. 4 Prediction performance on Alice 1A dataset

4.3 Results

Table 5 presents the simulations' results for the Alice 1A time series. SVM and LS-SVM obtained the lowest errors among the classical models, LSTM the lowest among DL models, eTS and ePL+ the lowest for the eFS, and GEN-NTSK (wRLS) and R-NTSK for the proposed fuzzy models.

Furthermore, eTS obtained the lowest NRMSE and NDEI among all simulations, and ePL+ the lowest MAPE, supporting the good performance of the eFS in real datasets. Although the eFS models secured the best overall metrics (eTS for NRMSE/NDEI and ePL+ for MAPE), the strength of our proposed models is validated by the statistical tests in Table 6. This analysis confirms that GEN-NTSK (wRLS)

Table 7 Simulations' results for the Alice 38 dataset

Model	NRMSE	NDEI	MAPE	Rules
<i>Classical models</i>				
KNN	0.23499	1.04722	0.37156	–
Regression Tree	0.21404	0.95385	0.34513	–
Random Forest	0.21363	0.95204	0.34780	–
SVM	0.21480	0.95723	0.34494	–
LS-SVM	0.20813	0.92755	0.33040	–
GBM	0.25371	1.13065	0.37142	–
XGBoost	0.22014	0.98103	0.34765	–
LGBM	0.21548	0.96026	0.34166	–
<i>DL</i>				
MLP	0.21231	0.94616	0.34199	–
CNN	0.20700	0.92250	0.32452	–
RNN	0.21020	0.93677	0.33125	–
LSTM	0.20858	0.92953	0.33252	–
GRU	0.21089	0.93981	0.32443	–
WaveNet	0.25446	1.13400	0.42215	–
<i>eFS</i>				
eTS	0.21114	0.94093	0.32321	3
Simpl_eTS	0.22844	1.01805	0.34967	36
exTS	0.20443	0.91105	0.33023	3
ePL	0.58321	2.59907	0.68757	1
eMG	0.28917	1.28866	0.42192	138
ePL+	0.25132	1.11999	0.35191	11
ePL-KRLS-DISCO	0.26151	1.16542	0.36657	33
<i>Discussed Models</i>				
NMR	0.24519	1.09269	0.39498	9
NTSK (RLS)	0.25395	1.13171	0.41610	1
NTSK (wRLS)	0.23656	1.05424	0.35931	11
GEN-NMR	0.26173	1.16638	0.35348	13
GEN-NTSK (RLS)	0.21581	0.96176	0.34244	1
GEN-NTSK (wRLS)	0.25224	1.12411	0.36122	13
R-NMR	0.21897	0.97582	0.33655	–
R-NTSK	0.21150	0.94252	0.33032	–
RF-NTSK	0.21087	0.93974	0.33716	–

Bold values denote the lowest error values observed for each class of model

held a statistically significant advantage over 8 of the 21 competing models, with R-NTSK outperforming 7. Regarding model complexity, a clear trade-off was observed. The simplest models, in terms of the number of rules, were ePL, NTSK (RLS), and GEN-NTSK (RLS), while eMG produced the most complex model. A qualitative analysis of the predictions, illustrated in Fig. 4, reveals a common challenge across all models: a difficulty in accurately forecasting extreme points. This suggests that the presence of outliers in the time series may be limiting the predictive performance of even the best-performing methods.

The simulation results for the Alice 38 time series, presented in Table 7, underscore the strong performance of the proposed ensemble models. Within the fuzzy model category, R-NTSK and RF-NTSK delivered the top results. Their performance was highly competitive against the best models from other classes, including LS-SVM, CNN, GRU, eTS, and exTS. The statistical analysis in Table 8 further substantiates these findings, confirming that R-NTSK held a statistically significant advantage over 8 of the competing models. In terms of rules complexity, ePL, NTSK (RLS), and GEN-NTSK (RLS) were the most parsimonious, generating the fewest rules, while eMG produced the most complex architecture. A visual inspection of the best predictions from each model class, presented in Fig. 5, allows for a qualitative comparison of their forecasting behavior in tracking the series' volatility.

For the Yulara 1 time series, the results in Table 9 show a competitive performance landscape, with the CNN model achieving the lowest overall error. Among the proposed methods, GEN-NMR and RF-NTSK delivered strong results, comparable to the top performers in other classes, such as RF, eTS, and exTS. The robust performance of RF-NTSK is further validated by the statistical analysis in Table 10, which indicates a significant performance advantage over 10 of the competing models. A notable finding was the extreme divergence in rules complexity; eMG created the model with 189 rules, whereas ePL, ePL+, NTSK (RLS), and GEN-NTSK (RLS) each produced a highly parsimonious model with just a single rule. Figure 6 offers a qualitative comparison of the predictive behavior of the best-performing model from each category, illustrating their different approaches to tracking the time series.

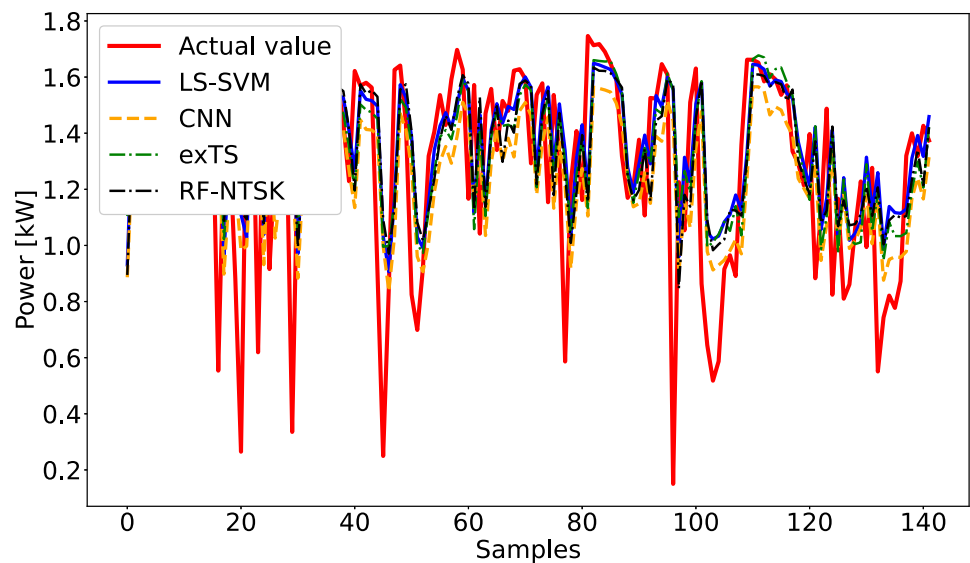
For the Yulara 5 time series, the results in Table 11 show a particularly strong performance from the proposed models, with R-NTSK achieving the lowest MAPE among all competing methods. While the evolving fuzzy model ePL-KRLS-DISCO secured the best overall NRMSE and NDEI, the proposed R-NTSK and GEN-NTSK (wRLS) models were highly competitive. Their performance is substantiated by the statistical analysis in Table 12, which confirms that R-NTSK held a statistically significant advantage over 11 of the compared models. In terms of number of rules, a notable finding was the high number of rules generated by some evolving systems, with Simpl_eTS producing 431 rules and eMG creating 372. Figure 7 provides a qualitative comparison of the forecasts from the best model in each class, highlighting their different strategies for tracking the series' dynamics.

4.4 Discussion

The experimental results provide a comprehensive evaluation of the proposed fuzzy models against a diverse suite of classical, DL, and eFS. Across the four time series datasets,

Table 8 Statistical test for Alice 38

Model	NMR	NTSK (RLS)	NTSK (wRLS)	GEN-NMR	GEN-NTSK (RLS)	GEN-NTSK (wRLS)	R-NMR	R-NTSK	RF-NTSK
KNN	0.87	0.99	0.55	0.91	0.01	0.87	0.09	0.01	0.00
Regression Tree	1.00	1.00	0.98	1.00	0.64	0.99	0.72	0.30	0.17
Random Forest	1.00	1.00	0.99	1.00	0.69	1.00	0.74	0.30	0.10
SVM	1.00	1.00	0.98	1.00	0.60	1.00	0.70	0.24	0.17
LS-SVM	1.00	1.00	1.00	1.00	0.99	1.00	0.94	0.90	0.88
GBM	0.30	0.51	0.11	0.69	0.00	0.47	0.01	0.00	0.00
XGBoost	0.99	1.00	0.97	1.00	0.24	0.99	0.44	0.04	0.02
LGBM	1.00	1.00	0.98	1.00	0.53	1.00	0.67	0.17	0.10
MLP	1.00	1.00	0.99	1.00	0.72	1.00	0.88	0.44	0.39
CNN	1.00	1.00	1.00	1.00	0.87	1.00	0.97	0.80	0.76
RNN	1.00	1.00	0.99	1.00	0.78	1.00	0.97	0.60	0.55
LSTM	1.00	1.00	1.00	1.00	0.86	1.00	0.96	0.72	0.68
GRU	1.00	1.00	0.99	1.00	0.71	1.00	0.89	0.54	0.50
WaveNet	0.12	0.48	0.13	0.63	0.00	0.45	0.01	0.00	0.00
eTS	1.00	1.00	0.99	1.00	0.76	1.00	0.84	0.52	0.48
Simpl_eTS	0.93	0.95	0.78	0.99	0.15	0.99	0.06	0.04	0.05
exTS	1.00	1.00	1.00	1.00	0.99	1.00	0.96	0.97	0.98
ePL	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
eMG	0.02	0.06	0.01	0.13	0.00	0.06	0.00	0.00	0.00
ePL+	0.40	0.54	0.22	0.69	0.08	0.52	0.06	0.05	0.05
ePL-KRLS-DISCO	0.18	0.35	0.06	0.51	0.00	0.30	0.00	0.00	0.00

Fig. 5 Prediction performance on Alice 38 dataset

the proposed models, particularly those enhanced with GA-based feature selection and ensembling, consistently ranked among the top-performing methods. Their predictive accuracy was highly competitive, often matching or exceeding that of complex DL architectures like CNNs and established methods like LS-SVM. For instance, on the Yulara 5 dataset,

R-NTSK achieved the lowest MAPE of all 29 competing models, demonstrating the framework's effectiveness.

While eFS like eTS and exTS also proved to be powerful predictors, the proposed models achieved their strong results with a non-evolving and conceptually simpler structure. This highlights a key advantage: the ability to attain high performance without the evolving nature and high number of rules

often associated with eFSs, such as the hundreds of rules generated by eMG and Simpl_eTS in several experiments.

Table 9 Simulations' results for the Yulara 1 dataset

Model	NRMSE	NDEI	MAPE	Rules
Classical Models				
KNN	0.21399	1.02255	0.39364	–
Regression Tree	0.21037	1.00523	0.28632	–
Random Forest	0.17592	0.84064	0.27256	–
SVM	0.21055	1.00610	0.34657	–
LS-SVM	0.21319	1.01869	0.36743	–
GBM	0.18105	0.86514	0.29454	–
XGBoost	0.20440	0.97670	0.30765	–
LGBM	0.17896	0.85515	0.28816	–
DL				
MLP	0.17511	0.83676	0.26667	–
CNN	0.16568	0.79168	0.25176	–
RNN	0.22641	1.08191	0.29806	–
LSTM	0.24267	1.15959	0.29554	–
GRU	0.22945	1.09639	0.29685	–
WaveNet	0.17783	0.84973	0.27339	–
eFS				
eTS	0.18750	0.89595	0.25613	4
Simpl_eTS	0.20622	0.98540	0.28118	45
exTS	0.16995	0.81211	0.25916	5
ePL	0.22578	1.07888	0.39487	1
eMG	0.28605	1.36686	0.51978	176
ePL+	0.21503	1.02751	0.36951	1
ePL-KRLS-DISCO	0.21213	1.01365	0.36978	20
Discussed Models				
NMR	0.27135	1.29663	0.38104	8
NTSK (RLS)	0.20606	0.98463	0.34618	1
NTSK (wRLS)	0.22261	1.06370	0.38723	2
GEN-NMR	0.19319	0.92315	0.26033	19
GEN-NTSK (RLS)	0.26105	1.24740	0.45110	1
GEN-NTSK (wRLS)	0.21812	1.04229	0.37640	19
R-NMR	0.26611	1.27157	0.36162	–
R-NTSK	0.20941	1.00066	0.35864	–
RF-NTSK	0.18351	0.87687	0.30299	–

Bold values denote the lowest error values observed for each class of model

A particularly noteworthy finding is the strong performance of the RLS-based models, which were often highly effective with just a single fuzzy rule. This outcome warrants deeper analysis. Since the RLS filter estimates a single, global parameter vector for the consequent, a one-rule configuration effectively collapses the fuzzy model into a global linear model. The excellent results from this single-rule model suggest that for these specific datasets, a well-tuned linear model

provides a very strong baseline and is sufficient to capture a significant portion of the underlying system dynamics.

However, the true value of the fuzzy framework is demonstrated by the wRLS-based models. These models, which use distinct local parameters for each fuzzy rule, consistently outperformed their RLS counterparts. This indicates that while a global linear model is effective, the fuzzy partitioning of the input space allows the wRLS models to capture additional nonlinearities and local behaviors that the global model misses, leading to statistically significant improvements in accuracy. Therefore, the framework not only produces a high-performing model but also provides insight into the degree of nonlinearity present in the data.

Beyond predictive accuracy, the proposed framework offers critical advantages in interpretability and transparency. The ability for a user to directly specify the number of rules provides explicit control over the accuracy–interpretability trade-off. This benefit is amplified by the GA-based feature selection, as exemplified by the GEN-NTSK (wRLS) model applied to the Alice 1A dataset. The GA selected an optimal subset of 6 out of the original 12 attributes, which delivered two simultaneous benefits:

- **Improved Interpretability:** The resulting rule base is far simpler and easier to analyze, as seen in Table 13. The antecedent of each rule is defined by only six features instead of twelve, focusing the analysis on the most influential variables. For even deeper transparency, the numerical fuzzy sets in the table could be translated into intuitive linguistic terms (e.g., 'low', 'medium', 'high').
- **Improved Accuracy:** The model's performance was enhanced by focusing only on the most relevant features. This demonstrates the GA's capacity to uncover complex correlations and filter out noisy or redundant attributes, leading to a more robust and effective model.

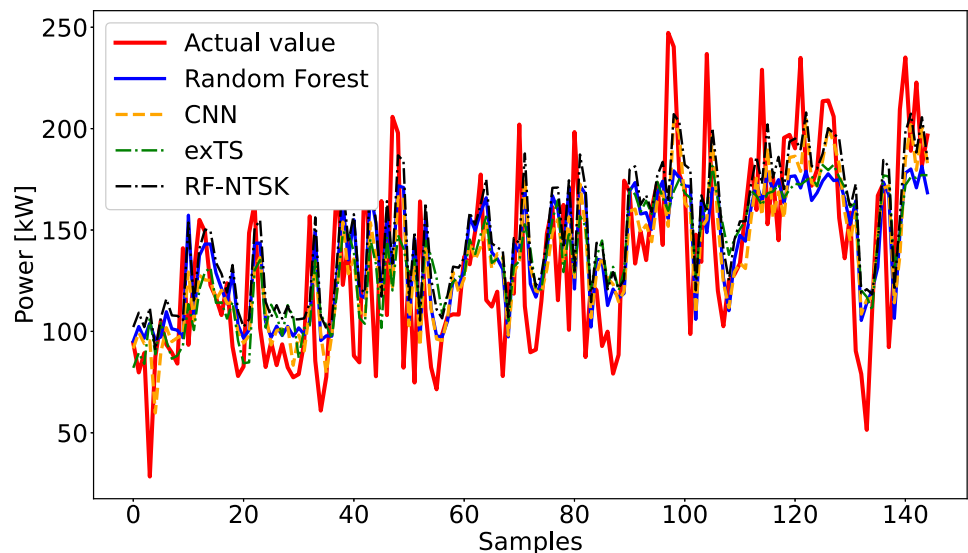
In summary, the simplicity of the rule induction mechanism, combined with the optimization from feature selection and the robustness added by ensembling, results in a framework that is both powerful and practical for real-world applications.

5 Conclusion

This paper addressed the persistent challenge of balancing accuracy and interpretability in machine learning, particularly in the context of complex real-world forecasting tasks. To overcome the "black-box" nature of DL and the limitations of traditional fuzzy systems, this work introduced and comprehensively evaluated NFISiS, a new family of data-driven FIS. The core contributions include the development of the NMR and the extension of the NTSK model, which were

Table 10 Statistical test for Yulara 1

Model	NMR	NTSK (RLS)	NTSK (wRLS)	GEN-NMR	GEN-NTSK (RLS)	GEN-NTSK (wRLS)	R-NMR	R-NTSK	RF-NTSK
KNN	1.00	0.21	0.79	0.06	1.00	0.65	1.00	0.33	0.00
Regression Tree	1.00	0.36	0.81	0.01	1.00	0.71	1.00	0.47	0.01
Random Forest	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	0.98
SVM	1.00	0.25	1.00	0.06	1.00	1.00	1.00	0.29	0.00
LS-SVM	1.00	0.12	1.00	0.05	1.00	1.00	1.00	0.00	0.00
GBM	1.00	1.00	1.00	0.91	1.00	1.00	1.00	1.00	0.63
XGBoost	1.00	0.56	0.90	0.10	1.00	0.84	1.00	0.65	0.02
LGBM	1.00	1.00	1.00	0.97	1.00	1.00	1.00	1.00	0.76
MLP	1.00	1.00	1.00	0.98	1.00	1.00	1.00	1.00	0.94
CNN	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
RNN	0.99	0.11	0.42	0.01	0.96	0.33	0.99	0.17	0.00
LSTM	0.93	0.02	0.16	0.00	0.81	0.11	0.93	0.04	0.00
GRU	0.99	0.08	0.36	0.01	0.94	0.27	0.99	0.13	0.00
WaveNet	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	0.93
eTS	1.00	0.94	0.99	0.75	1.00	0.98	1.00	0.95	0.36
Simpl_eTS	1.00	0.50	0.84	0.15	1.00	0.77	1.00	0.58	0.05
exTS	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99
ePL	0.99	0.00	0.01	0.01	1.00	0.00	0.98	0.00	0.00
eMG	0.25	0.00	0.00	0.00	0.04	0.00	0.18	0.00	0.00
ePL+	1.00	0.03	1.00	0.04	1.00	0.85	1.00	0.00	0.00
ePL-KRLS-DISCO	1.00	0.24	0.86	0.06	1.00	0.74	1.00	0.37	0.00

Fig. 6 Prediction performance on Yulara 1 dataset

systematically enhanced with wrapper-based feature selection using GA (GEN-NMR and GEN-NTSK) and ensemble techniques (R-NMR, R-NTSK, and RF-NTSK).

The proposed models were rigorously validated on challenging real-world PV energy forecasting datasets. Experimental results demonstrated that the enhanced models, particularly GEN-NTSK and RF-NTSK, achieved highly

competitive and often superior performance compared to a wide array of benchmarks. Notably, these models rivaled the accuracy of established eFS and complex DL architectures while offering a significantly simpler and more transparent structure. This indicates that integrating feature selection and ensemble methods effectively enhances model robustness

Table 11 Simulations' results for the Yulara 5 dataset

Model	NRMSE	NDEI	MAPE	Rules
<i>Classical Models</i>				
KNN	0.23408	1.10290	0.33711	–
Regression Tree	0.21492	1.01262	0.29218	–
Random Forest	0.21306	1.00388	0.28702	–
SVM	0.22354	1.05322	0.29446	–
LS-SVM	0.20783	0.97920	0.28330	–
GBM	0.21769	1.02570	0.28887	–
XGBoost	0.22848	1.07649	0.30705	–
LGBM	0.21589	1.01717	0.28938	–
<i>DL</i>				
MLP	0.21564	1.01601	0.30108	–
CNN	0.21186	0.99821	0.28915	–
RNN	0.29585	1.39393	0.37098	–
LSTM	0.30403	1.43246	0.37513	–
GRU	0.22070	1.03985	0.28804	–
WaveNet	0.21745	1.02455	0.29647	–
<i>eFS</i>				
eTS	0.21658	1.02046	0.28506	3
Simpl_eTS	0.22494	1.05983	0.29851	431
exTS	0.20669	0.97386	0.28372	3
ePL	0.20992	0.98907	0.29031	1
eMG	0.32813	1.54603	0.41719	372
ePL+	0.20917	0.98555	0.28964	4
ePL-KRLS-DISCO	0.20501	0.96595	0.28255	20
<i>Discussed Models</i>				
NMR	0.32884	1.54935	0.39332	2
NTSK (RLS)	0.22291	1.05029	0.31533	1
NTSK (wRLS)	0.21398	1.00818	0.28825	5
GEN-NMR	0.29698	1.39928	0.36355	15
GEN-NTSK (RLS)	0.22080	1.04035	0.29857	1
GEN-NTSK (wRLS)	0.20625	0.97176	0.28503	15
R-NMR	0.36554	1.72230	0.42459	–
R-NTSK	0.20632	0.97209	0.27714	–
RF-NTSK	0.20749	0.97761	0.27978	–

Bold values denote the lowest error values observed for each class of model

and predictive power, especially on noisy, high-dimensional data.

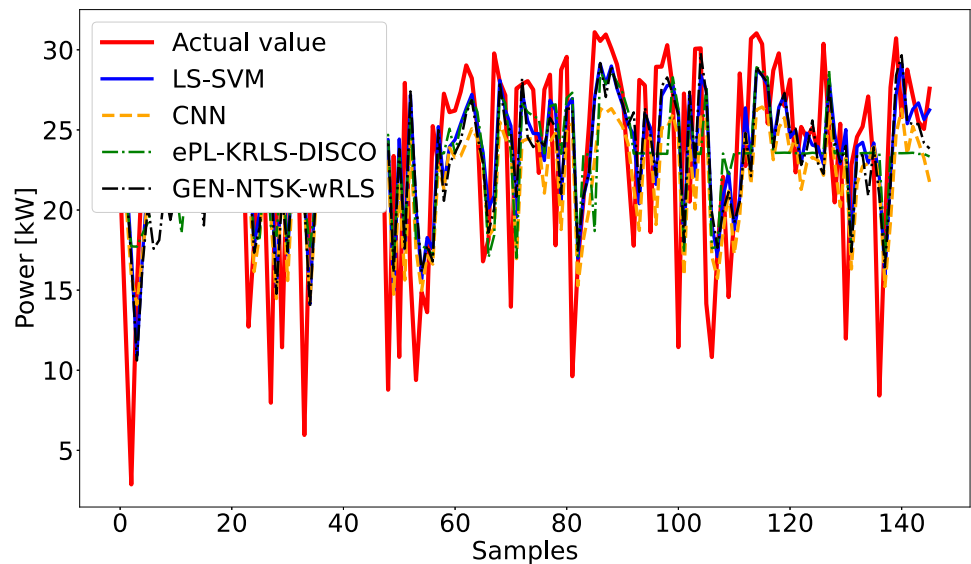
The primary advantage of the proposed framework lies in its unique combination of autonomy, performance, and interpretability. Unlike many conventional rule-based models, this approach grants users direct control over the number of rules, allowing them to explicitly manage the trade-off between model interpretability and predictive performance. The use of a GA for feature selection further optimizes performance while enhancing transparency by identifying the most critical predictive attributes. These characteristics render the proposed framework particularly suitable for critical

applications, such as renewable energy management, where transparent and reliable decision-support tools are essential.

Based on these findings, several promising avenues for future research can be identified. The model architecture could be enhanced by exploring Type-2 fuzzy systems to better manage high levels of uncertainty, investigating alternative membership functions beyond Gaussian, and integrating advanced algorithms like Kernel Recursive Least Squares (KRLS) for more robust consequent parameter estimation. The learning paradigm could also be extended to include mechanisms for online learning, enabling the models to adapt sequentially from data streams. Finally, the scope of evaluation could be broadened through a formal computa-

Table 12 Statistical test for Yulara 5

Model	NMR	NTSK (RLS)	NTSK (wRLS)	GEN-NMR	GEN-NTSK (RLS)	GEN-NTSK (wRLS)	R-NMR	R-NTSK	RF-NTSK
KNN	1.00	0.04	0.02	1.00	0.04	0.00	1.00	0.00	0.00
Regression Tree	1.00	0.79	0.43	1.00	0.81	0.08	1.00	0.03	0.02
Random Forest	1.00	0.82	0.57	1.00	0.88	0.13	1.00	0.07	0.01
SVM	1.00	0.47	0.05	1.00	0.27	0.00	1.00	0.00	0.00
LS-SVM	1.00	0.98	0.93	1.00	1.00	0.26	1.00	0.23	0.46
GBM	1.00	0.68	0.27	1.00	0.67	0.05	1.00	0.02	0.00
XGBoost	1.00	0.28	0.03	1.00	0.14	0.00	1.00	0.00	0.00
LGBM	1.00	0.73	0.38	1.00	0.74	0.08	1.00	0.05	0.02
MLP	1.00	0.72	0.42	1.00	0.70	0.12	1.00	0.13	0.13
CNN	1.00	0.85	0.66	1.00	0.92	0.14	1.00	0.11	0.08
RNN	1.00	0.00	0.00	0.59	0.00	0.00	1.00	0.00	0.00
LSTM	1.00	0.00	0.00	0.08	0.00	0.00	1.00	0.00	0.00
GRU	1.00	0.58	0.09	1.00	0.51	0.01	1.00	0.00	0.00
WaveNet	1.00	0.68	0.28	1.00	0.68	0.03	1.00	0.03	0.02
eTS	1.00	0.73	0.29	1.00	0.77	0.03	1.00	0.01	0.04
Simpl_eTS	1.00	0.43	0.11	1.00	0.32	0.02	1.00	0.01	0.02
exTS	1.00	0.99	0.97	1.00	1.00	0.39	1.00	0.44	0.58
ePL	1.00	0.94	0.82	1.00	0.97	0.09	1.00	0.15	0.27
eMG	0.51	0.00	0.00	0.14	0.00	0.00	0.91	0.00	0.00
ePL+	1.00	0.95	0.84	1.00	0.96	0.22	1.00	0.27	0.37
ePL-KRLS-DISCO	1.00	0.97	0.92	1.00	0.99	0.58	1.00	0.60	0.69

Fig. 7 Prediction performance on Yulara 5 dataset**Table 13** Rules of GEN-NTSK (wRLS) for Alice 1A with four rules

Rule	Diffuse radiation [W/m^2]	Rainfall [mm]	Wind Direction [$^\circ$]	Current [A]	Power [kW]	Global radiation [W/m^2]	Next power [kW]
1	57.67 (27.35)	0.07 (0.20)	24.22 (18.12)	5.04 (0.85)	2.48 (0.49)	301.66 (69.41)	$[-2.56, -1.37]$
2	61.40 (37.21)	0.56 (2.41)	31.20 (15.20)	4.48 (1.07)	2.18 (0.64)	264.34 (88.11)	$[-1.37, -0.18]$
3	54.35 (35.91)	0.46 (2.45)	31.33 (15.58)	4.54 (1.11)	2.24 (0.57)	270.44 (78.31)	$[-0.18, 1.01]$
4	79.98 (34.09)	0.96 (3.43)	27.92 (15.82)	3.82 (1.33)	1.79 (0.76)	218.76 (95.23)	$[1.01, 2.20]$

tional complexity analysis and by testing the framework on a more diverse range of challenging datasets to further validate its scalability and generalization capabilities.

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Data Availability To promote reproducibility, the models are provided in an open-source library named nfisis (<https://pypi.org/project/nfisis/>), and the datasets used in this study are publicly available at <https://doi.org/10.5281/zenodo.17179229>.

Declarations

Ethical and informed consent for data used Fulfilled

Competing interests The authors declare no competing interests.

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